Overview

Wiki *Enhanced dissipation for passive scalars*

9 November 2026

Discuss on Zulip: #overview, topic "Overview".

A passive scalar is a quantity carried by a fluid without acting on it, such as heat, a dye in water or a pollutant in the air. If f is its concentration, u the velocity of an incompressible flow, and $\nu > 0$ the molecular diffusivity, then f solves the advection–diffusion equation

$$\partial_t f + u \cdot \nabla f = \nu \Delta f, \qquad \nabla \cdot u = 0.$$

Diffusion alone dissipates the scalar on a time scale of order $1/\nu$, which is very long when ν is small. The flow, on the other hand, dissipates nothing: it only rearranges the scalar. In doing so, it can transfer large-scale structures to small scales, for instance by stretching the scalar into thin filaments, and at small scales diffusion is much more efficient. The result is that f decays on a time scale much shorter than $1/\nu$: this is *enhanced dissipation*.

Beyond passive scalars. The same mechanism, acting on the vorticity of a two-dimensional fluid, helps keep shear flows stable at large Reynolds numbers, against perturbations that are small enough in terms of the Reynolds number; the page Why it matters expands on this.

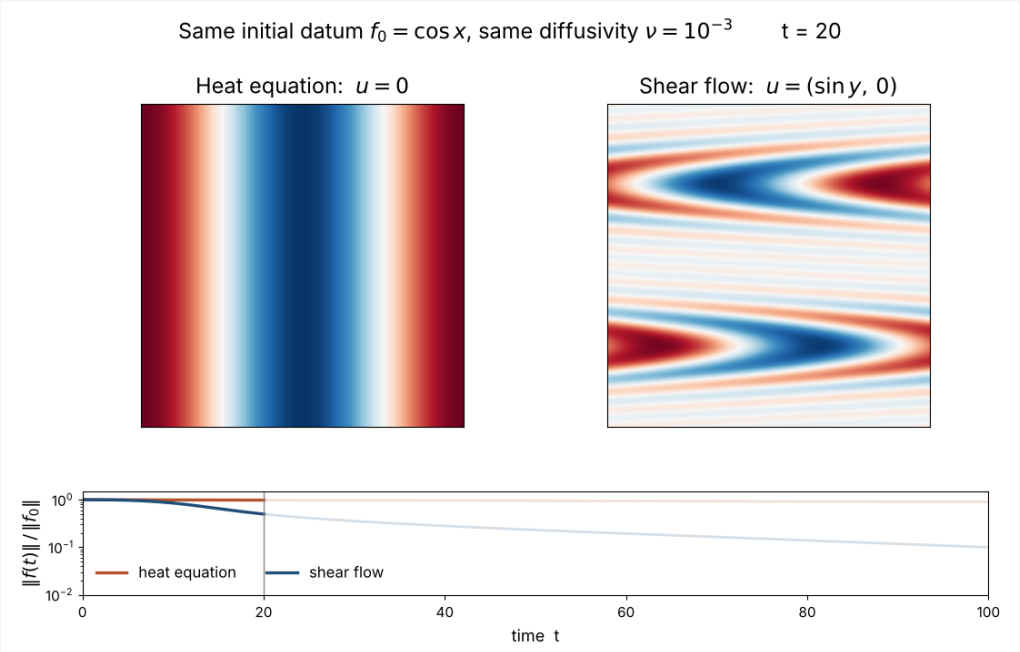
How, and by how much, does an incompressible velocity field speed up the dissipation of a diffusing scalar?

The two videos below show the phenomenon in the simplest setting, a periodic box in two dimensions. In the first, a steady shear flow significantly accelerates the dissipation of vertical stripes, orthogonal to its streamlines.

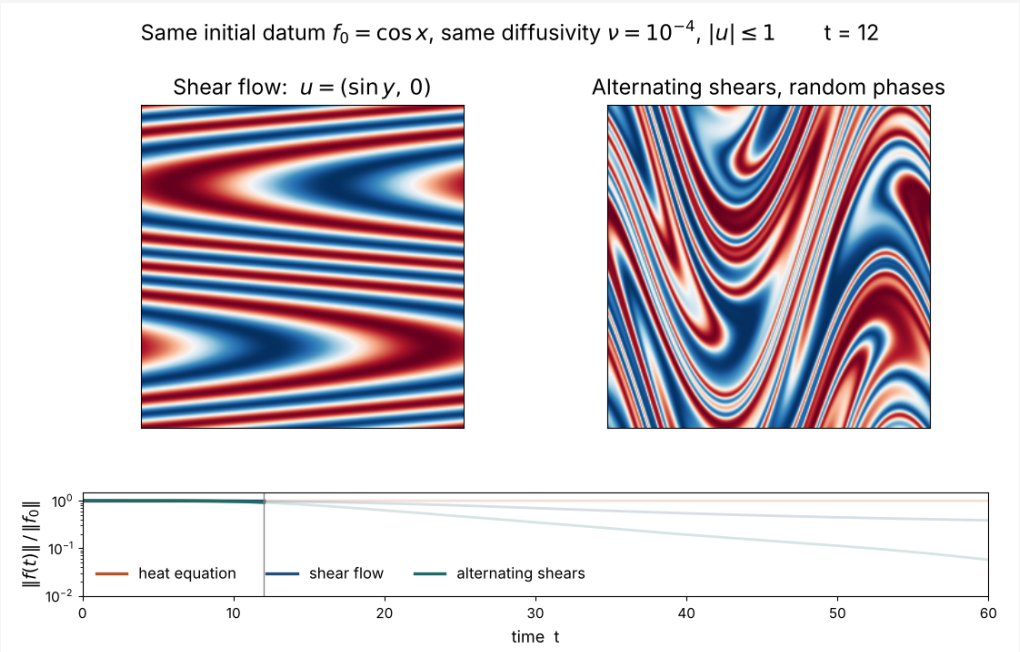
The second video compares two flows of the same size. A steady shear stretches the scalar only along its own streamlines. Shears that alternate in direction, with random phases (the model of Pierrehumbert), stretch and fold the scalar, which then dissipates much faster and more uniformly in space.

Quantifying the time scale of enhanced dissipation is the subject of this wiki. Laminar flows, such as shear flows, typically dissipate the scalar on time scales of order ν^{-a} with $0 < a < 1$, whereas *mixing* flows, such as the alternating shears, can dissipate it on time scales of order $|\log \nu|$. The questions are which time scales are possible, which flows achieve them, and how one proves it.

Mixing: the problem without diffusion. When $\nu = 0$ the equation is pure transport: the flow conserves every L^p norm of f and dissipates nothing. It can still *mix*, that is, move the scalar to ever smaller scales, so that f converges weakly to its mean while its L^2 norm stays constant. The speed of mixing can be measured by the decay of a negative Sobolev norm, such as $\|f\|_{H^{-1}}$. Mixing is a field of its own, with its own questions and methods; this wiki covers only its relation to enhanced dissipation.



Video. Left: the heat equation. Right: the same initial datum carried by the shear flow $u = (\sin y, 0)$, with the same diffusivity. Bottom: the size of the two solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record ***num-heat-vs-shear****.*



Video. Left: the shear flow $u = (\sin y, 0)$. Right: shears that alternate, horizontal and vertical, each acting for 3 units of time, with random phases. Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-4}$; both velocity fields have size at most 1. Bottom: the size of the solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record ***num-alternating-shears****.*

The setting, made precise. In most of the wiki, the domains without boundary are the torus $\mathbb{T}^2$ and the periodic strip $\mathbb{T} \times \mathbb{R}$; some problems are set in the whole plane $\mathbb{R}^2$. The domains with boundary are the disk and the periodic channel $\mathbb{T} \times [0, 1]$; the boundary conditions are stated where they are used. The initial datum f_0 is in L^2 and has mean zero; for shear and circular flows, it also has zero average along each streamline, since the flow does not act on that average. The size of a solution is its L^2 norm. One definition of the dissipation time of a flow is the time it needs to reduce every initial datum by a fixed factor, say one half; the flow enhances dissipation when this time is much shorter than $1/\nu$. Definitions and notation are collected in Notation.

1 Map of the wiki

- *Motivation and history*: why the problem matters and where it comes from.
- *Techniques through examples*: enhanced dissipation for two flows, Couette and Poiseuille, proved once with each method, so that the methods can be compared on the same example.
- *Techniques and general results*: the general theorem behind each method, with its proof, for spectral methods, hypocoercivity, stochastic methods and Hörmander theory.
- *Extensions and recent advances*: the state of the art, related results and open problems.
- *Toolkit*: the notation and the standard analysis that the proofs use.
- *Numerical methods*: the codes behind the pictures.
- *Lean formalization*: the statements checked by a proof assistant.
- *About this wiki*: how to contribute and who did what.

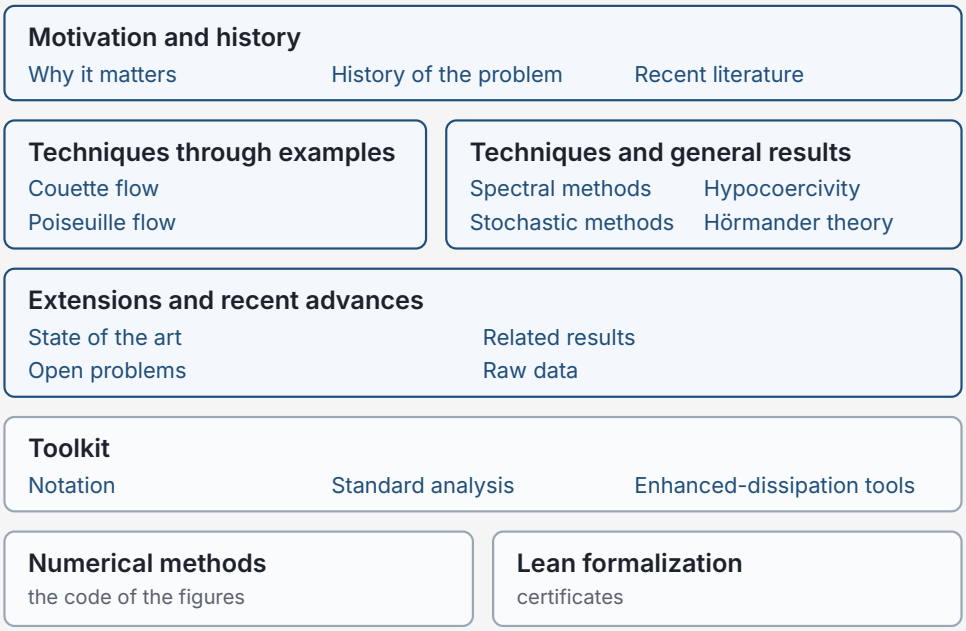


Diagram. The parts of the wiki; click a name to open its page.